

1. (4 pts) What effect does inserting the iris inside the cavity have on the lasing? Explain what happens when the aperture diameter is reduced. Why does this happen?
2. (4 pts) Sketch the intensity patterns of the TEM_{30} , TEM_{21} , and TEM_{12} rectangular modes.

 TEM_{30} TEM_{21} TEM_{12}

3. (4 pts) You used a glass plate held at Brewster's angle to polarize the laser beam. Calculate the Brewster's angle for glass. Assume $n = 1.5$ for the glass index of refraction.
4. (4pts) Explain how the Brewster angle plate causes the beam to be polarized? How did you tell it was polarized?
5. (4 pts) If a thin, straight wire is positioned directly in the center of the axis of the HeNe laser inside the cavity, which of the modes – TEM_{00} , TEM_{10} , TEM_{20} , and TEM_{12} -- would you expect might lase? Explain.